Module (5)-Assignment

Submitted by: Khansa Rubab

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Materials and Method

Variant Calling, Effect Prediction, and Genomic Distribution Analysis

Genomic DNA was utilized from an ethically approved Breast cancer samples **SRA run accession** SRR33505761 were mapped against the Homo sapiens chromosomes 19 reference (NC_000019) collected with ethical approval, and genomic DNA was extracted using standard protocols are assemble to specific contigs. Variant calling was conducted using Geneious Mapper v9.0.5, with parameters set as follows: a minimum variant frequency of approximately 0.25, a maximum variant P-value of approximately 10^-5, and a minimum strand-bias P-value of approximately 10^-5. Variants with a strand bias exceeding 65% were investigated for their association with the reference sequence.

Furthermore, predicted effects of identified variants were determined using bioinformatics tools to assess potential functional consequences, such as missense mutations or splice site alterations. This analysis was crucial for understanding the impact of variants on gene expression, protein function, and cellular processes relevant to breast cancer pathogenesis.

Furthermore, genomic distribution analysis of single nucleotide polymorphisms (SNPs) and insertions/deletions (InDels) was performed by determining their frequency within 100-kilobase (kb) intervals on the Breast cancer samples. This method provided insights into the spatial distribution of genetic variants across the genome, facilitating a comprehensive understanding of their potential roles in disease development and progression. By elucidating the genomic landscape of Breast malignancy and its implications on gene expression and cellular functions, this study aimed to explore the impact of non-synonymous SNPs in protein-coding genes on leukemia susceptibility and progression, seeking biomarkers for targeted therapeutics in breast cancer treatment.